

Buyside Quant Prep Roadmap

NewWhale Career

Each module mapped to the decks that cover it

1. Probability and Brainteasers

Q03-Q08, Q12, Q13

Classical probability, combinatorics (dice, coins, cards), Bayes, expectation and variance, Markov chains and gambler's ruin, uniform / geometric / normal distributions, symmetry arguments.

Every quant seat. Market makers test this live and fast.

2. Statistics: Estimation and Inference

Q10 (survey), Q21 (deep)

Bias / variance / MSE of estimators, method of moments, MLE and its properties, laws of total expectation and variance, confidence intervals, hypothesis testing, min / max of random variables.

HF research seats; the warm-up layer before regression questions.

3. Statistics: Regression and Model Building

Q19, Q20 (the core pair)

OLS from objective to normal equations, all six assumptions and their failure modes, t / F inference, R-squared traps, multicollinearity and VIF, bias-variance, cross-validation (incl. walk-forward), gradient descent, ridge / lasso / elastic net.

The highest-yield technical area for hedge fund and buyside interviews.

4. Options Theory

S09-S10 (foundations), Q22 (Greeks), Q09 (stochastic calculus)

Payoffs and put-call parity, Black-Scholes formula and assumptions, implied volatility and the smile, and how every Greek behaves deep OTM / ATM / deep ITM and as expiry approaches.

Options desks and vol funds; Q09 only for researcher / pricing seats.

5. Programming

Q14, plus daily practice outside the slides

Python first (C++ later, only after Python is solid). OOP, lists / tuples / dicts / heaps, hashmaps, big-O, searching and sorting, pandas data manipulation, mutable vs immutable. Practice easy-to-medium algorithm problems until 20-minute solves feel routine.

Screens and pair-coding rounds at every systematic shop.

6. Applied Projects

QM series (hands-on modeling track)

Build one real, validated strategy you can defend for 20 minutes. Project menu: market-beta (CAPM) study, GARCH volatility forecast, pairs trade with cointegration test, gradient-boosted return classifier, factor-model replication.

The differentiator: interviews end up spending most time here.

7. Question Bank and Mocks

Q16 (easy), Q17 (medium), Q18 (hard), Q15 (fit), Q02 (mental math)

112 interview problems with full solutions. Work them on paper before reading answers. Finish with timed mock interviews.

The last two weeks of prep should mostly live here.

The 8-Week Plan

NewWhale Career

A realistic schedule at 5 to 8 focused hours per week; compress to 4 weeks only if full-time

Weeks 1-2	Probability core: Q03 (fundamentals), Q04 (expected value), Q05 (counting), Q06 (Bayes). Ten easy bank questions per week. Five minutes of mental math (Q02) daily.
Week 3	Distributions and processes: Q07, Q08. Classic puzzles Q13. Keep the easy bank running.
Week 4	Statistics I: Q10 as the map, then Q21 in depth. Derive the Bernoulli and Normal MLEs on paper without notes.
Week 5	Statistics II: Q19 then Q20, the highest-yield week of the plan. Reproduce the OLS derivation and the ridge / lasso contrast on paper.
Week 6	Options: S09 / S10 if new to options, then Q22 until you can rebuild both Greek tables unprompted. Researcher-track candidates add Q09.
Week 7	Programming: Q14 patterns, daily easy / medium problems in Python, one pandas exercise per day. Start the medium bank (Q17).
Week 8	Integration: finish Q17 / Q18, one applied project write-up (QM track), Q15 behavioral prep, and at least one timed mock.

What each seat tests

Market makers and prop shops

Mental math under time pressure, probability and EV games, market-making intuition (Q11), game theory (Q12). Speed and composure matter as much as correctness.

Hedge funds (fundamental or pod)

Regression and model building (Q19 / Q20), statistics, a defensible project, and market awareness. Expect 'here is a dataset, how would you find a signal?'

Quant research at systematic funds

Everything above plus stochastic calculus (Q09) and deeper ML discussion. Your project write-up carries the most weight here.

Rule of thumb: an interviewer can take any answer one level deeper. Prep one level deeper than the answers you plan to give.